Enterprise Disaster Recovery & Backup Architecture

1. Project Overview

Objective:
Achieve Recovery Point Objective (RPO) of 1 Hour and Recovery Time Objective (RTO) of 4 Hours.

Scope:
- EC2 Workload Backup
- Shared File Storage (EFS)
- Cross-Region Disaster Recovery
- Hybrid Integration

2. Step 1: Install Required Packages

NFS client packages are required to mount Amazon EFS on EC2 instances.
Commands:
sudo apt update
sudo apt install nfs-common –y

```
ubuntu@ip-10-0-15-71:~$ sudo apt update
sudo apt install nfs-common -y
Hit:1 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
40 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-10-0-15-71:~$
```

3. Step 2: EBS Multi-Attach Setup

Created an EBS volume with gp3 type and size 20GB. Enabled Multi-Attach and attached it to two EC2 instances.

Commands:
lsblk
sudo mkfs -t ext4 /dev/nvme1n1
sudo mount /dev/nvme1n1 /data

Verified no downtime during volume resize.

```
buntu@ip-10-0-7-229:~$ sudo mount /dev/nvme1n1 /task3
buntu@ip-10-0-7-229:~$ df -h
ilesystem                                          Size  Used Avail Use% Mounted on
dev/root                                           6.8G  2.1G  4.7G  31% /
mpfs                                               456M     0  456M   0% /dev/shm
mpfs                                               183M  912K  182M   1% /run
mpfs                                               5.0M     0  5.0M   0% /run/lock
fivarfs                                            128K  3.9K  120K   4% /sys/firmware/efi/efivars
dev/nvme0n1p16                                     881M   94M  726M  12% /boot
dev/nvme0n1p15                                     105M  6.2M   99M   6% /boot/efi
mpfs                                                92M   12K   92M   1% /run/user/1000
s-0f43f53e6f00e617e.efs.ap-south-2.amazonaws.com:/  8.0E     0  8.0E   0% /home/ubuntu/task3
dev/nvme1n1                                         98G   24K   93G   1% /task3
buntu@ip-10-0-7-229:~$
```

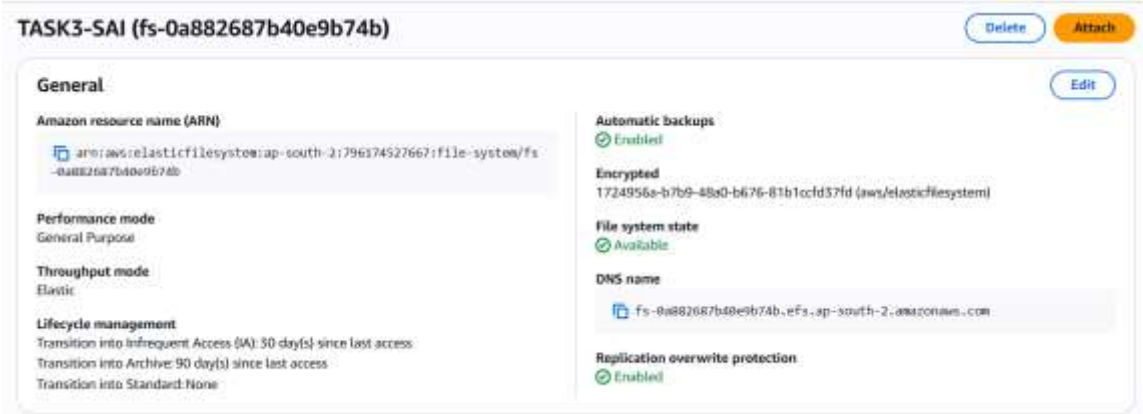
4. Step 3: EFS Setup

Created an Amazon EFS file system and enabled mount targets in multiple Availability Zones.

Command:
sudo mount -t nfs4 fs-xxxx:/ /mnt/efs

```
ubuntu@ip-10-0-7-229:~$ sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport
t fs-0f43f53e6f00e617e.efs.ap-south-2.amazonaws.com:/ task3
ubuntu@ip-10-0-7-229:~$ df -h
Filesystem                                         Size  Used Avail Use% Mounted on
/dev/root                                          6.8G  2.1G  4.7G  31% /
tmpfs                                              456M     0  456M   0% /dev/shm
tmpfs                                              183M  916K  182M   1% /run
tmpfs                                              5.0M     0  5.0M   0% /run/lock
efivarfs                                           128K  3.9K  120K   4% /sys/firmware/efi/efivars
/dev/nvme0n1p16                                    881M   94M  726M  12% /boot
/dev/nvme0n1p15                                    105M  6.2M   99M   6% /boot/efi
tmpfs                                               92M   12K   92M   1% /run/user/1000
fs-0f43f53e6f00e617e.efs.ap-south-2.amazonaws.com:/  8.0E     0  8.0E   0% /home/ubuntu/task3
ubuntu@ip-10-0-7-229:~$
```

Lifecycle Policies:
- Move to Infrequent Access after 30 days
- Move to Archive after 90 days

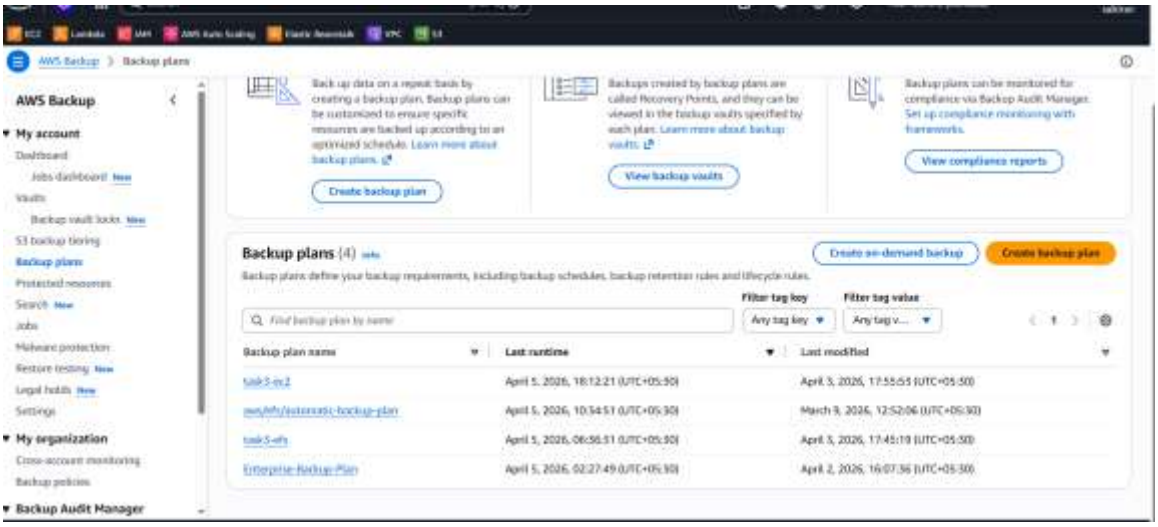


5. Step 4: AWS Backup Configuration

Configured AWS Backup with a backup vault using KMS encryption.

Backup Plan:
- EBS: Hourly backups, 24-hour retention
- EFS: Daily backups, 30-day retention
- AMI: Daily backups, 30-day retention

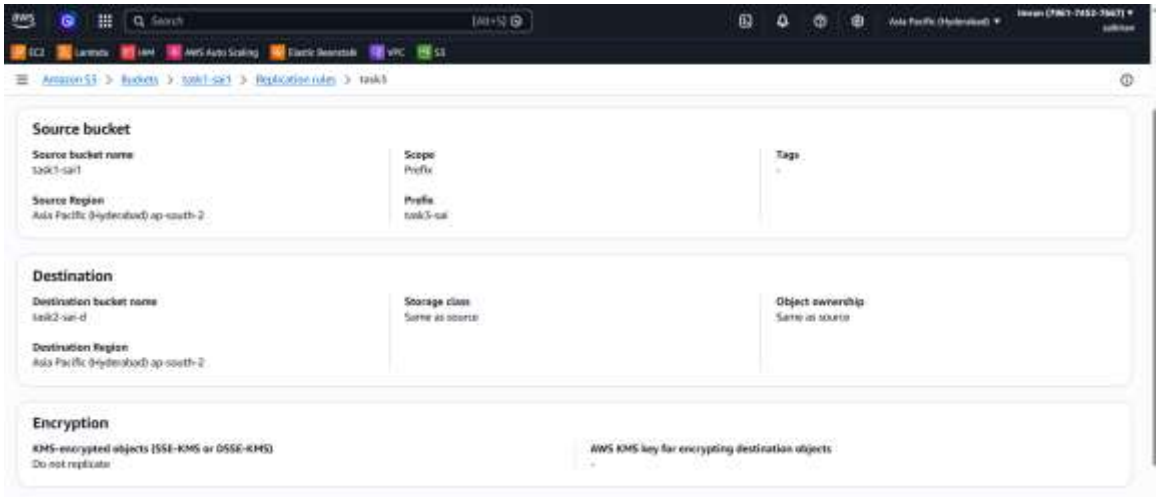
Cross-Region Copy:
- Destination: us-west-2
- Retention: 1 Year



6. Step 5: S3 Cross-Region Replication (CRR)

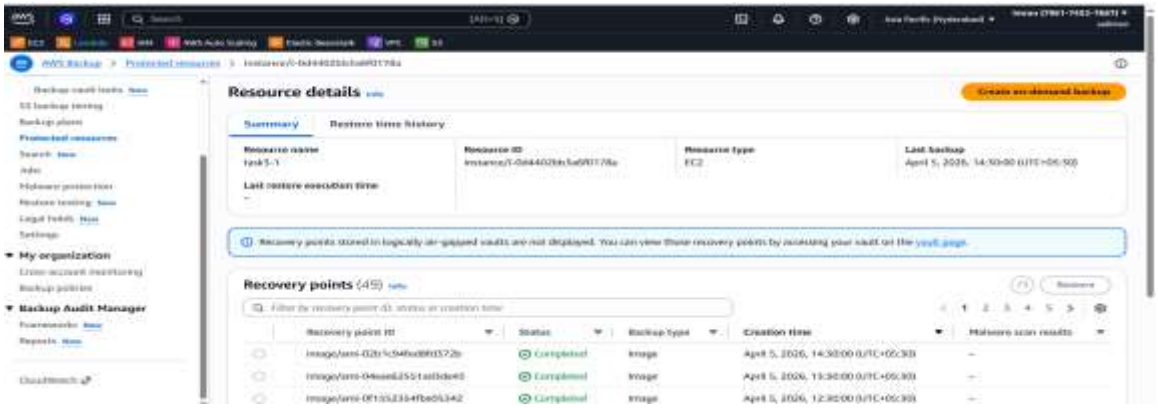
Created source and destination S3 buckets in different regions. Enabled versioning and configured CRR.

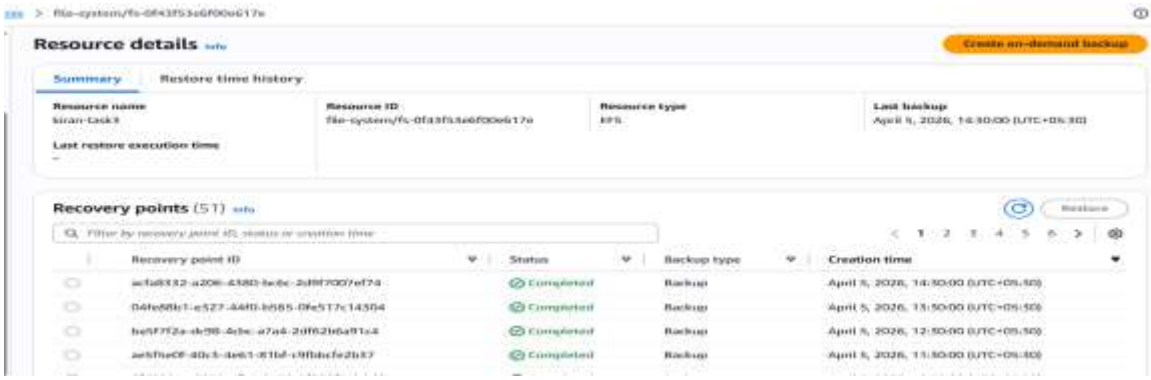
Validation:
Uploaded data in the source bucket and verified replication in the destination bucket.



7. Validation & Testing

Backup Validation:
- Triggered manual backup from AWS Backup
- Confirmed successful job completion





-demand backups.

, status, resource ID, resource type or message category

< 1 2 3 4 5 6 7 ... 31 >

Resource ID	Resource type	Creation time	Start by
volume/vol-0c191be40e17eed86	EBS	April 5, 2026, 17:30:00 (UTC+05:30)	April 6, 2026, 01:30:00 (UTC+05:30)
task1-sai1	S3	April 5, 2026, 17:30:00 (UTC+05:30)	April 6, 2026, 01:30:00 (UTC+05:30)
task2-sai-d	S3	April 5, 2026, 17:30:00 (UTC+05:30)	April 6, 2026, 01:30:00 (UTC+05:30)
volume/vol-048a828755a7b6f45	EBS	April 5, 2026, 17:30:00 (UTC+05:30)	April 6, 2026, 01:30:00 (UTC+05:30)
volume/vol-0c38b72510482742d	EBS	April 5, 2026, 17:30:00 (UTC+05:30)	April 6, 2026, 01:30:00 (UTC+05:30)
volume/vol-04accc56effbc6fca	EBS	April 5, 2026, 16:30:00 (UTC+05:30)	April 6, 2026, 00:30:00 (UTC+05:30)
task2-sai-d	S3	April 5, 2026, 16:30:00 (UTC+05:30)	April 6, 2026, 00:30:00 (UTC+05:30)
volume/vol-07cc575d03ee8f6e1	EBS	April 5, 2026, 16:30:00 (UTC+05:30)	April 6, 2026, 00:30:00 (UTC+05:30)
volume/vol-048a828755a7b6f45	EBS	April 5, 2026, 16:30:00 (UTC+05:30)	April 6, 2026, 00:30:00 (UTC+05:30)
task1-sai1	S3	April 5, 2026, 16:30:00 (UTC+05:30)	April 6, 2026, 00:30:00 (UTC+05:30)

8. Conclusion

This architecture successfully meets business requirements by ensuring:

- High availability
- Data durability
- Disaster recovery readiness
- Compliance with RPO (1 hour) and RTO (4 hours)